Vandit Parekh

Mobile: +91-9157585969

Email: parekhvandit@gmail.com

Education

- **Vellore Institute of Technology, Bhopal** — Bhopal, India
  Bachelor's of Technology - Computer Science; — *September 2023 -*

Courses: *Operating Systems, Data Structures, Analysis Of Algorithms, Problem Solving and Programming, Object Oriented Programming, Computer Architecture, Database Management System, Numerical Methods, Differential Calculus, Calculus and Laplace Transformation, Principles of Management and Organizational Behavior, Fundamentals in AI and ML*

Skills Summary

- **Languages:** C++, Python, Java, SQL, HTML Unix/Linux, Assembly
- **Platforms:** Linux, Windows
- **Co-Curricular:** Public Speaking, Technical Writing, Team Collaboration, Debate

Projects

- **Automated E-TOLL Plaza**:
  - Spearheaded the development of an intelligent, automated toll system that eliminates congestion by implementing a seamless electronic toll collection (ETC) mechanism.
  - Collaborated with a team of 2 to design a future-oriented E-TOLL system integrated with an automatic payment mobile application.
  - Implemented graph theory algorithms to compute the most cost-effective, time-efficient routes from point A to point B based on real-time traffic and toll data.
  - Engineered an automated payment system by integrating a robust Payment API for hassle-free transactions.
  - Enhanced security by implementing encryption protocols for user data and financial transactions.
  - Reduced waiting times at toll booths by an estimated 70 percent, improving traffic flow and reducing fuel wastage.
  - Successfully deployed a prototype in a simulated environment, demonstrating the project's viability for large-scale smart city implementation.
  - Implemented rigorous testing protocols using simulations and real-world field tests to ensure system robustness.
  - Utilized advanced analytics to monitor toll usage patterns, optimizing route recommendations and overall performance.

- **PosePerfect: Real-Time Yoga Pose Estimation & Correction Game**:
  - Developed **PosePerfect**, an interactive application that helps users perform yoga poses correctly by providing real-time feedback and scoring—each correct pose awards one point.
  - Integrated OpenCV for video capture, MediaPipe for full-body pose estimation, and a deep learning model (via Keras) to classify and evaluate yoga poses in real time.
  - Engineered a responsive Pygame interface featuring custom fonts, animated backgrounds, interactive buttons, and real-time overlays for reliable pose tracking.
  - Designed a multi-stage game flow with home, start, and gameplay screens, offering dynamic feedback and challenges to enhance user performance.
  - Integrated user analytics to track pose accuracy trends over time, enabling personalized feedback and progress monitoring.
  - Tested and refined the system across multiple devices and lighting conditions to ensure robust performance in diverse real-world settings.
  - Co-authored a research paper titled *Performance Assessment of Body Pose Estimation in Rehabilitation Settings*, evaluating MediaPipe's efficacy in tracking joint movements during rehabilitation and yoga exercises.
  - Overall, PosePerfect not only aids yoga enthusiasts in perfecting their poses but also contributes valuable insights into pose estimation for rehabilitation.

Achievements

- Secured **top 50** in the **HealthHack Hackathon** among 850+ teams, showcasing strong problem-solving, algorithmic thinking, and efficient coding.
- Serving as **Chief of Operations** at VITBMUN. Oversaw content creation, facilitated discussions, handled logistics, managed finances and led the orgainisin team